

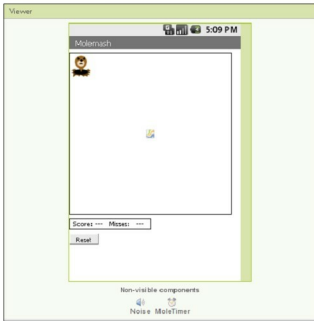


Figure 3: The viewer, informally a 'design canvas'

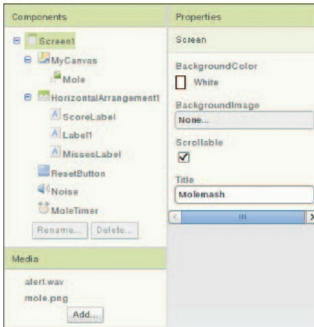


Figure 4: The components tree and properties pane

running on the phone (or in an emulator) within a few seconds. This is an exhilarating way to learn, try, test, and debug. It's impressive how quickly you can get significant functionality working in half an hour, which with conventional coding would have taken an average programmer an afternoon for a single 'code, package, install-on-the-phone and test' cycle.

Let's now look at the respective roles of the AIA Web application and the Blocks Editor, before going on to the phone or emulator, which is (merely) used to provide a live

run of the app you are developing.

The AIA Web application

It's time we looked at some images to illustrate what we're talking about. I took some screen captures with the fourth official Google tutorial project (a game called MoleMash) opened, to point out some elements which didn't occur in previous tutorial projects.

Figure 1 is an 'overview' of the whole AIA Web application. As you will notice, the application has the *My Projects*, *Design* and *Learn* links in the grey area at the top. The first two switch between a list of your existing projects and the currently-opened project in the design area. The third, *Learn*, leads to a page with links to *Getting Started*, *Tutorials*, *Reference Documentation*, *FAQ*, *Troubleshooting* and more.

The dark green bar below that contains (from left to right) the project name, *Save*, *Save As* and the *Checkpoint* button. This last appears to create a copy of the project with the name you specify; so it's not initially obvious how different it is from the *Save As* button, other than that it appends 'checkpoint_n' to the project name. To the right of the page, in this same green bar, are buttons to open the Blocks Editor, and to package the currently open project as an installable '.apk' file, which you can download to your PC, or install directly to the connected phone.

Components



Note: The term 'component' refers to an item available in the AIA Web application's palette, which you drag onto the design viewer/canvas to use in your application. If you have done GUI programming before, these components are conceptually equivalent to widgets or controls in other GUI tool-kits.

Figure 2 shows a close-up of the palette of components, with the Basic group expanded. The App Inventor team has created plenty of components for use with an Android phone, including:

- The basic UI components with the familiar *Button*, *CheckBox*, *Image*, *Label*, *TextBox*, *PasswordTextBox*, and *ListPicker* components. There are also *Canvas* (for drawing) and *Clock* (for timer events) components.
- The database component to store information: Your applications can even store data created by you in a database, using a *TinyDB* component. You could create a make-a-quiz application, in which teachers can save quiz questions for students to answer later, perhaps something like a flash-card memory-building application, or a game.
- Components for screen layout management: Containers that can contain other controls, and arrange them vertically, horizontally, or in a table.
- Components for simple animation, including an image